

FlowPrecision: Advancing FPGA-Based Real-Time Fluid Flow Estimation with Linear Quantization

Tianheng Ling, Julian Hoeber, Chao Qian, Gregor Schiele

Intelligent Embedded Systems Lab, [University of Duisburg-Essen](#) 47057 Duisburg, Germany
{tianheng.ling, chao.qian, gregor.schiele}@uni-due.de, julian.hoeber@stud.uni-due.de

Abstract—In industrial and environmental monitoring, achieving real-time and precise fluid flow measurement remains a critical challenge. This study applies linear quantization in FPGA-based soft sensors for fluid flow estimation, significantly enhancing Neural Network model precision by overcoming the limitations of traditional fixed-point quantization. Our approach achieves up to a 10.10% reduction in Mean Squared Error and a notable 9.39% improvement in inference speed through targeted hardware optimizations. Validated across multiple data sets, our findings demonstrate that the optimized FPGA-based quantized models can provide efficient, accurate real-time inference, offering a viable alternative to cloud-based processing in pervasive autonomous systems.

Index Terms—Flow Estimation, Soft Sensors, In-situ AI, Quantization, FPGA-based Inference, Embedded Systems

I. INTRODUCTION

Pervasive in-situ AI systems have immense potential in various applications, such as industrial and environmental monitoring. One critical task in these settings is accurately measuring fluid flow, which is challenging, particularly in harsh environments such as drilling pipes or sewers. Direct-contact physical sensors often degrade quickly and require frequent maintenance, while non-contact sensors may lack precision or may be cost-prohibitive. An emerging solution to this challenge is to use *soft sensors*, which utilize data from cheap, non-contact sensors to estimate flow rate accurately.

Recent advancements in flow estimation have utilized physical formulas, statistical models, and deep learning algorithms. Among them, Tomperi et al. [1] employed multivariate linear regression with cost-efficient distance sensors for wastewater flow estimation. Chhantyal et al. [2] integrated data fusion with Artificial Neural Networks (ANNs) for ultrasonic level sensor-based flow estimation. Noori et al. [3] combined shallow ANNs with radar sensors, and Liang et al. [4] developed models for complex flow scenarios using ultrasonic sensors. However, these approaches primarily focus on cloud-based soft sensors, susceptible to timing and network dependency issues, potentially affecting service reliability.

In response to the need for autonomy and reliability in soft sensor deployment, our study shifts the focus to deploying ANN-based soft sensor algorithms directly on Internet of Things (IoT) devices. This local analysis approach mitigates network dependencies but introduces new challenges due to the limited computational resources available on IoT devices. These constraints necessitate the optimization of ANN models for efficient execution, a process we address through model

quantization. Specifically, we propose quantizing ANN models for deployment on embedded Field Programmable Gate Arrays (FPGAs) using linear quantization with adaptive scalars. This method aligns quantization parameters with the actual data distributions of each tensor, thereby enhancing model precision and balancing energy efficiency. Our key contributions include:

- We apply linear quantization to Multi-Layer Perceptron (MLP) models for fluid flow estimation, reducing test loss by up to 10.10%.
- We conduct an ablation study to validate our approach against precision loss issues in fixed-point quantization.
- We test our method across three data sets, demonstrating its broad applicability and robustness.
- We develop a specialized hardware accelerator that enhances model precision while maintaining energy efficiency and acceptable resource utilization overhead.

The remainder content is organized as follows: Section II reviews pertinent literature, setting the stage for our research. Section III delves into the technical aspects of fixed-point and linear quantization. Section IV details our approach and implementation. Section V describes the experimental setup, and Section VI presents and discusses our findings. Finally, Section VII concludes the paper and outlines future research.

II. RELATED WORK

Accurate flow estimation using soft sensors requires high-bit quantization to ensure precision, mainly due to its regression-oriented nature. Ling et al. [5] advanced in this area by quantizing an MLP model for FPGA platforms, employing a uniform power-of-2 scaling factor in their 8-bit fixed-point quantization approach. While their approach showcased FPGA deployments' speed and energy efficiency, it faltered in precision when compared to linear quantized models on Microcontroller Units (MCUs). The primary limitation of fixed-point quantization stems from its assumption of uniform tensor distribution, leading to notable precision loss amidst real-world data variability. In our pursuit to enhance model precision for soft sensor applications on FPGAs, we also examined the work of Jain et al. [6]. Their introduction of per-tensor scaling considerably ameliorated compatibility across diverse hardware platforms and more effectively adapted to the variability inherent in tensor distributions. Beyond refining the fixed-point quantization, linear quantization is a formidable alternative. This technique applies a zero point and scaling factor to convert floating-point tensors into integer tensors linearly.